

restart:

$$eq1 := CrO2 + Cr2O3 = 1$$

$$CrO2 + Cr2O3 = 1 \quad (1)$$

$$eq2 := \frac{Cr2O3}{CrO2} = 0.35$$

$$\frac{Cr2O3}{CrO2} = 0.35 \quad (2)$$

$$solve(\{eq1, eq2\}, \{CrO2, Cr2O3\})$$

$$\{Cr2O3 = 0.2592592593, CrO2 = 0.7407407407\} \quad (3)$$

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$$eq1 := Cr + Cr2O3 + CrO2 = 1$$

$$Cr + Cr2O3 + CrO2 = 1 \quad (4)$$

$$eq2 := \frac{Cr2O3}{CrO2} = \text{constant}$$

$$\frac{Cr2O3}{CrO2} = \text{constant} \quad (5)$$

$$solve(\{eq1, eq2\}, \{Cr, Cr2O3, CrO2\})$$

$$\{Cr = 1 - \text{constant } CrO2 - CrO2, Cr2O3 = \text{constant } CrO2, CrO2 = CrO2\} \quad (6)$$

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$$eq1 := CrO2o + CrO2e = 1$$

$$CrO2o + CrO2e = 1 \quad (7)$$

$$eq2 := \frac{CrO2o}{CrO2e} = 2$$

$$\frac{CrO2o}{CrO2e} = 2 \quad (8)$$

$$solve(\{eq1, eq2\}, \{CrO2o, CrO2e\})$$

$$\left\{CrO2e = \frac{1}{3}, CrO2o = \frac{2}{3}\right\} \quad (9)$$